

WEEK 5 – SPL 5

Pure metals – ductile and soft

This softness arises because of their **low lattice resistance** — that is, there are few obstacles in the crystal structure to resist dislocation motion (which leads to plastic deformation).

Therefore, to make the ‘stronger’ we use:

Method	Mechanism	Process	Microstructure Effect	Typical Application
Solid Solution Strengthening	Solute atoms distort the crystal lattice, hindering dislocation motion	Add alloying elements; atoms dissolve into solid solution (e.g., Zn in Cu)	Uniform solid solution, lattice strain due to atom size mismatch	Brass (Cu-Zn), Stainless steel (Cr in Fe)
Precipitation Hardening	Fine precipitates form and obstruct dislocations	Heat treat to form solid solution, quench, then age to form precipitates	Precipitates dispersed in the matrix (e.g., CuAl_2 in Duralumin)	Al-Cu (Duralumin), Ni-based superalloys
Work Hardening	Plastic deformation increases dislocation density, which entangles	Cold working (e.g., rolling, drawing, hammering)	High dislocation density and strain-hardened regions	Cold-rolled steel, metal forming processes

Solid solution – substitutional (big) and interstitial (small)

METAL TYPES

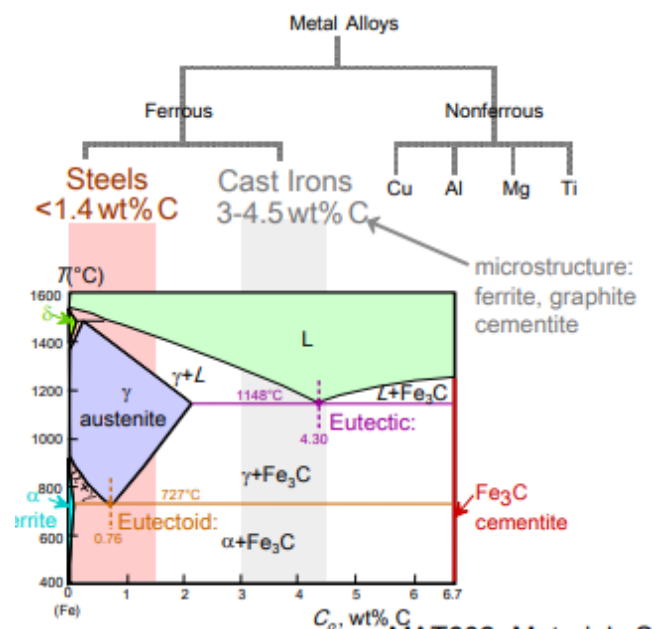
FERROUS ALLOYS – IRON IS THE PRIME CONSTITUENT- low carbon steel, high carbon steel, stainless steel, cast iron, grey iron and white iron.

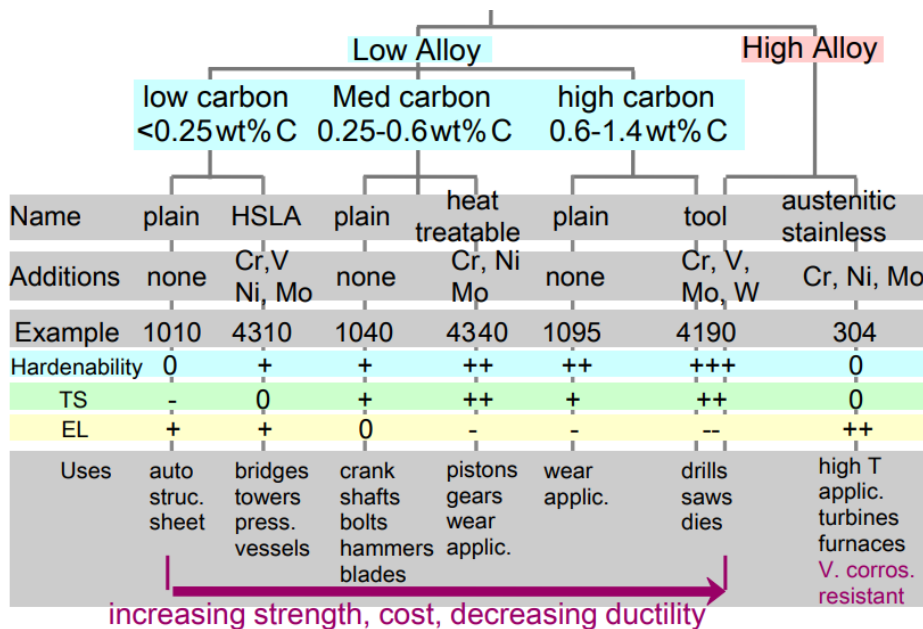
NON- FERROUS ALLOYS – copper alloys, aluminium alloys, titanium alloys and magnesium alloys.

TAXONOMY OF METALS

Anything below 1.4% carbon = steels- can plastically deform

Anything above = cast iron





Types of cast iron

1. Gray Cast Iron

- Graphite in flakes (sharp, irregular shapes)
- Flakes act as stress concentrators → weak under tension
- But it's stronger under compression and offers:
 - Excellent vibration damping (used in machinery bases)
 - Good wear resistance
- Matrix = ferrite (softer) or pearlite (stronger)

2. Ductile (Nodular) Iron

- Made by adding Mg or Ce → changes graphite to nodules (spheres)
- Spheroidal graphite means less stress concentration → much better ductility and toughness
- Matrix is often pearlitic for strength

3. White Cast Iron

- No graphite; carbon is mostly iron carbide (Fe_3C or cementite)
- Very hard and wear-resistant, but very brittle
- Formed via fast cooling, doesn't allow graphite to form

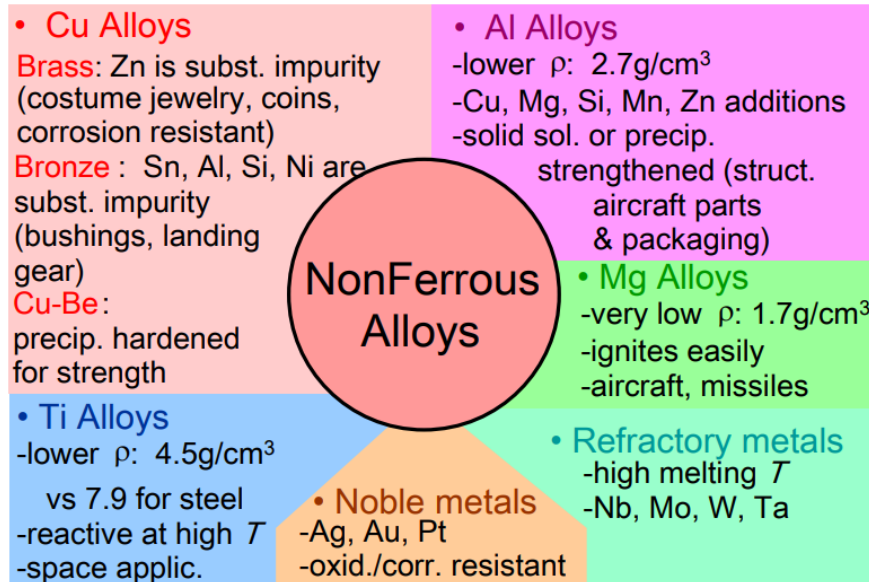
4. Malleable Cast Iron

- Made by heat treating white iron at 800–900°C for 30+ hours
- This transforms iron carbide into graphite "rosettes"
- Offers good ductility with moderate strength

NON- FERROUS

Nonferrous alloys are metallic alloys that do **not contain iron (Fe)** as a major component. These materials are often valued for:

- **Lightweight**
- **Corrosion resistance**
- **High strength-to-weight ratio**
- **Specialized thermal or electrical properties**



Cost depends on melting as all materials need to be melted to be processed

DESIGN CONSIDERATION

Design Challenge	Explanation
Cooling Shrinkage	Metals contract when cooled from melt, which affects dimensional accuracy.
Air Gap Formation	Shrinkage during cooling creates a gap between metal and mould, reducing heat transfer.
Work Hardening	Plastic deformation increases strength, limiting further shaping unless annealed.
Limited Plastic-Forming	Excessive deformation risks cracking; shaping must be done in steps.
Corners Difficult to Form	Metals struggle to fill sharp mould corners, leading to defects.
Unidirectional Pressing	Forging presses push in one direction; liquids can fill complex moulds better.
Phase Changes	Structural changes on heating/cooling influence mechanical properties and process control.

INGOT CASTING- usually reheated/reshaped into useful products – is the first stage

- Ingot casting involves **pouring molten metal into molds to form simple, solid shapes called ingots.**
- The **melting temperature of a material determines the casting process and mold design.**
- **Metals must be heated above this temperature to be cast**, and mold materials must withstand it without degrading.

Ingot CASTING

Metal	Melt temp/°C
Tungsten	3400
Silica	1900
Platinum	1770
Iron	1536
Nickel	1453
Mild steel	1413
Gold	1063
Copper	1083
Aluminium	660
Magnesium	650
Zinc	420
Lead	327

- Processes depend on the materials produced.
- Mould type depends on material and hence melt temperature.
- Tabulated are the minimum casting temperatures for different metals /°C.

MAT602- Materials Selection in Design

Q: What are typical mold characteristics for ingot casting?

A: Molds are generally **simple in shape with rounded corners**. This design eases removal of the solidified ingot and minimizes stress concentrations that might lead to cracks during cooling.

Q: What is meant by the grain structure in metals?

A: After solidification, metals develop a **crystalline grain structure**. The nature of this structure affects mechanical properties like strength and toughness.

Q: What causes the contraction cavity at the top of the ingot?

A: As molten metal cools, it **shrinks**. Since solidification starts at the mold walls and proceeds inward and upward, a **cavity often forms at the top**, where the last part remains molten and contracts as it solidifies.

Q: What happens to impurities during solidification?

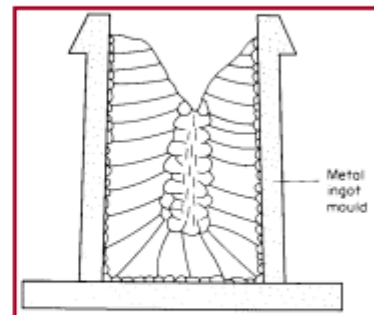
A: Impurities are usually expelled from the solidifying front and **concentrated in the last melt zones**, which often end up at the top of the ingot.

Q: How do gas bubbles form during ingot casting?

A: Gases (like nitrogen) that were dissolved in the molten metal can come out of solution upon cooling, forming **gas bubbles or porosity** in the ingot.

Ingot Casting

- Moulds are usually simple shapes with rounded corners to facilitate removal.
- Metals have a grain structure.
- Shrinkage creates a contraction cavity at the top of the ingot.
- Impurities are first expelled from the solid and concentrated in the last melt zones.
- Similarly, gas bubbles may form from dissolved nitrogen being released on cooling.



MAT602- Materials Selection in Design

PROBLEMS WITH INGOT CASTING

1. Contraction cavities

- Ingot casting creates a **shrinkage hole** at the top because metal shrinks when it cools.

2. Segregation of impurities

- Impurities (and alloying elements) are **pushed to the last melt zones**, usually the center or top, creating **non-uniform composition**.

3. Gas porosity

- Gases like nitrogen can get trapped and form **bubbles or voids**.

4. Larger starting size

- Ingots are large, so more **energy and effort** is needed to shape them later (e.g., into sheets or rods).

Therefore, **CONTINUOUS CASTING**

1. No shrinkage cavity

2. Less segregation:

- The columnar grains (crystals) form over **shorter distances**, so impurities don't have time to separate much — leading to **more uniform material**.

3. Smaller product = less work later:

- The output is already in **slab or rod shape**, so **less rolling or forging is needed** later to get it into usable form.

Continuous Casting:

- Uses a **permanent mould** (like a water-cooled copper mould).
- The molten metal flows continuously, solidifies, and exits as a solid bar or sheet.
- There is **no expendable pattern or mould** — it is **reused over and over**.
- Used for producing **slabs, billets, and blooms** (like in steel mills).

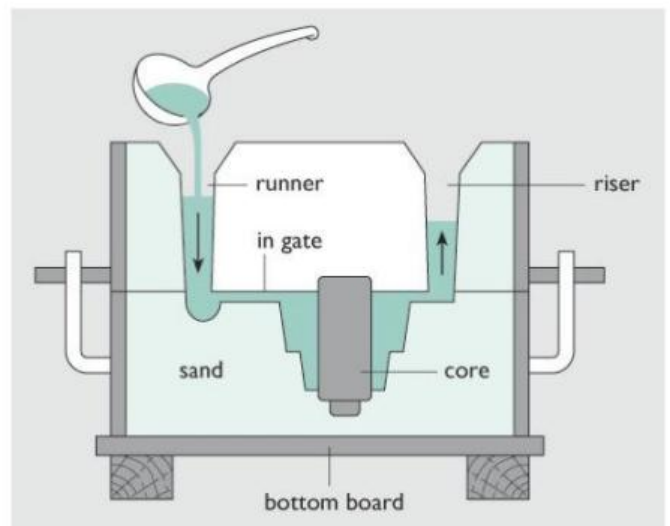
PERMANENT PATTERN (SAND CASTING)

Sand casting is a method where **molten metal is poured into a mold made of sand**. Once the metal cools and solidifies, the sand mold is **broken to remove the final part**.

- A **pattern** is made (usually wood, metal, or plastic).
- It is pressed into sand to make a mold cavity.
- The pattern is **removed**, leaving behind the shape.
- A **core** may be added for internal features.
- Molten metal is **poured into the mold** via the gate.
- Once it cools and solidifies, the sand is **broken apart**, and the casting is removed.
- The **mold is discarded**, but the **pattern is reused** to make more castings.

Why use this method?

- **Low cost** for small production runs
- **Flexible shapes** (including complex geometries)



- **Large parts** can be made easily
- Reusable pattern makes it **cost-effective** over many castings

Automated sandcasting

Automated sand casting uses **machines** to rapidly and repeatedly **form sand molds**, enabling **high-volume production** of cast parts. This method dramatically increases casting speed compared to traditional hand-packed molds.

DIE CASTING / PERMANENT MOULDS

Die casting involves injecting **molten metal under pressure** into a **reusable metal mould (die)** to produce accurately shaped parts. Unlike sand casting, the mould is **not destroyed** after each use — it's **opened and reused** for thousands of cycles.

High-temperature materials needed for the die:

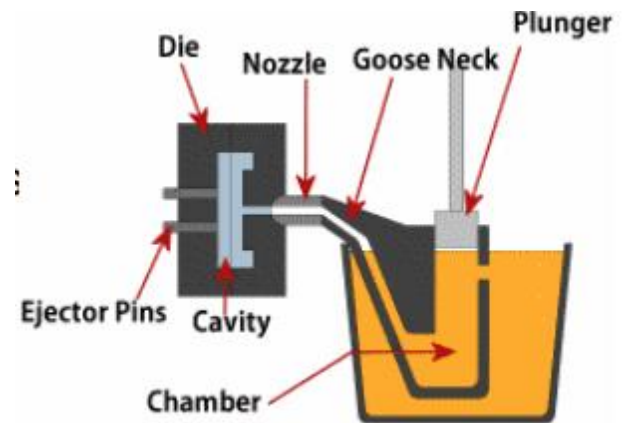
- Since molten metal is hot, the die must be made of materials like **hardened tool steel** that can withstand **repeated heating**.

Limits the process to lower melting point metals:

- Dies cannot handle extremely high temperatures (like for steel), so this process is mostly used with **aluminium, zinc, magnesium, etc.**

High cost of moulds (dies):

- Dies are **expensive** to manufacture due to the **complex design, precision machining, and materials**.
- This is why die casting is **economical only when**:
 - Producing **large quantities** (to spread the mould cost),
 - Making **small components** (less metal = less heat = longer die life).

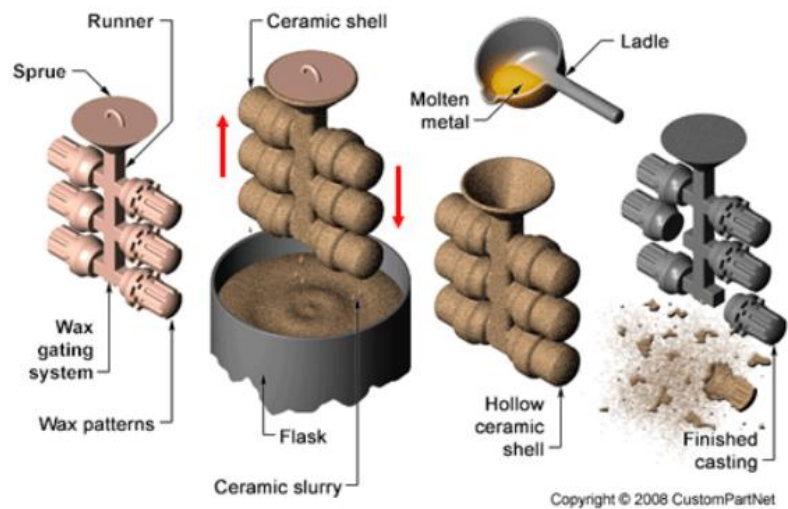


Expendable Mould (Investment Casting)

1. Making a **wax model** of the part
2. Coating it in **ceramic** to form a mould
3. Melting away the wax (hence "lost wax")
4. Pouring molten metal into the hollow ceramic shell
5. Breaking the shell to reveal the final part

Expendable Mould (Investment Casting)

Pattern Tree Shell-Making Investment Casting Casting



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MAT602- Materials Selection in Design

FORMING – METAL FABRICATION METHODS

Process	Force Type	Product Type	Notes
Forging	Compressive	Tools, shafts	High strength; hot or cold
Drawing	Tensile	Wires, tubes	Requires clean, lubricated dies
Rolling	Compressive	Sheets, rails, beams	Hot = easy forming, Cold = better finish
Extrusion	Compressive/Ram	Long rods, hollow tubes	Ductile metals, continuous profile

FORGING Method: A metal blank is placed between dies and hit with a hammer or press.

DRAWING Method: A metal rod is pulled through a **die**, reducing its cross-sectional area.

ROLLING Method: A metal piece is passed between two rotating rolls.

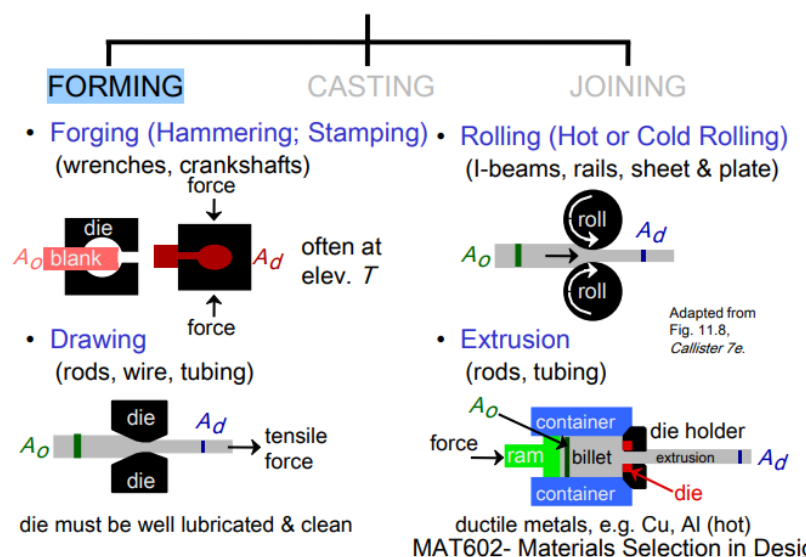
EXTRUSION Method:

- A metal **billet** is placed in a chamber.
- A **ram** pushes it through a **die**.
- The metal flows out as a continuous profile.

METAL FORMING

We use metal forming because casting has defects – cant fill voids/ cavity and holes therefore cost can be reduced
FORMING BETTER IMPROVED MECH PROPERTIES

Metal Fabrication Methods - Forming



ductile metals, e.g. Cu, Al (hot)
MAT602- Materials Selection in Design

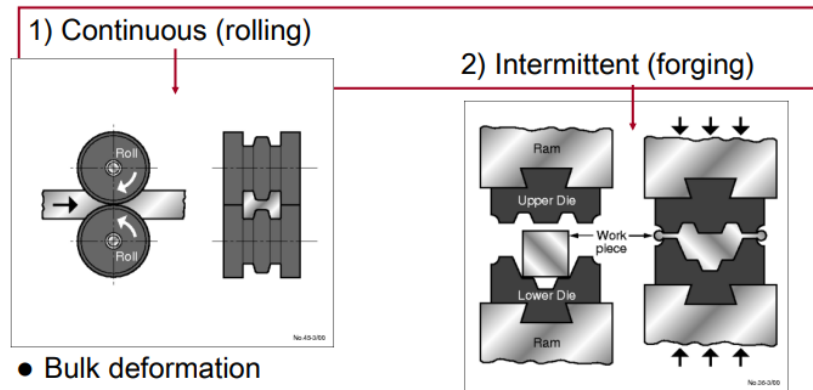
Long shapes harder to cast

Best casting from EUTECTIC ALLOYS but not always desirable

INHOMOGENITY- common in casting but plastic deformation reduces this. Casting often causes **uneven structure** due to cooling. Forming evens it out through **plastic deformation**.

Some materials difficult to process as liquid due to high MELT temp – REFRACTORY MATERIALS(NIOBIUM AND TUNGSTEN) - better shaped through forming or powder processes.

CLASSIFICATIONS OF FORMING PROCESSES



- Bulk deformation
- This involves tri-axial stresses.
- Starting material has a high volume to surface area.
- Examples forging in open or closed dies.

FORGING

Hot Forging

- **Benefits:**
 - Allows **large deformations** (big changes in shape)
 - Easier to shape metals due to reduced strength at high temperatures
- **Drawbacks:**
 - Poorer **surface quality**
 - Lower **dimensional accuracy**
 - Problems like **oxidation** (metal reacting with air) and **warpage** (distortion when cooling)

Cold Forging

- **Benefits:**
 - Produces **more precise parts**
 - Results in a **better surface finish**
- **Drawbacks:**
 - Requires **higher pressures** to deform metal (since it's not softened by heat)
 - Limited deformation due to **work hardening** (material becomes stronger and harder as it's shaped)

PROCESS	COLD OR HOT?	REASON
DRAWING	☑ Cold forming	Performed at room temperature
HOT FORGING	✗ Hot forming	Involves heating metal above recrystallization temp
SAND CASTING	✗ Not a forming process	It's a casting process
WELDING	✗ Joining process	Not forming at all

1. Formability and Bend Radius

- **Formability** is how easily a sheet can be bent or stretched without cracking.
- It's often measured as the **minimum bend radius** in multiples of sheet thickness (**t**):
 - **t = 1** → excellent (can bend sharply)
 - **t = 4** → average (needs wider bend)

2. Radii Design Rule

- Bending edges and corners should have **large radii**.
- Ideally, radius $\geq$ sheet thickness to **prevent cracking** during bending.

3. Stretching & Drawing Limits

- Formability also affects how far a sheet can be **stretched or deep drawn** (e.g. into a cup shape).
- Poor formability leads to **necking** (local thinning) and **failure**.

4. Forming Limit Diagrams (FLDs)

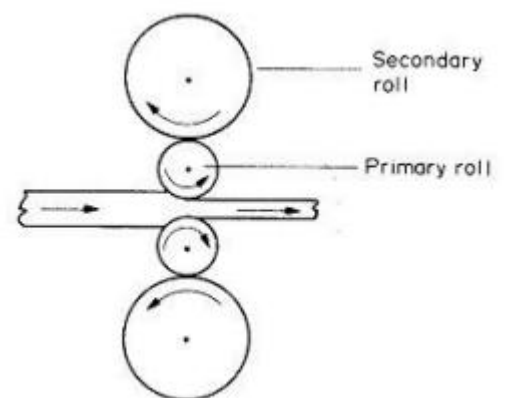
- FLDs provide a **quantitative guide** to forming safety.
- They plot **strain combinations** (major vs. minor strain) that a sheet can survive before failing.
- Used in **advanced simulations** and product design to predict failure points.

5. Design Strategy

- Engineers design components to ensure **strain levels stay below the forming limit**, as shown in FLDs.
- This prevents tearing, cracks, or part rejection during manufacturing.

Rolling

Feature	Hot Rolling	Cold Rolling
Temperature	High ($T > 0.35 T_m$)	Low ($T < 0.35 T_m$)
Pressure	Low	High
Surface Finish	Rough	Smooth
Strength	Low	Higher (due to work hardening)
Accuracy	Less	More



Well designed rolling mills have rolls of **small diameter (primary roller)** these can be supported by additional secondary roles which do not touch the work piece!!

METAL CONSIDERATIONS

 **Left: Idealized (Perfectly Plastic) Material**

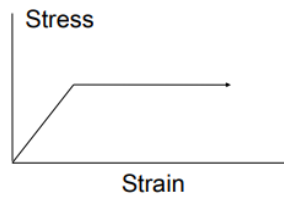
 **Right: Real Metal Behavior (with Work Hardening)**

- Stress increases again after yielding due to **work hardening**.
- As the metal is deformed, it becomes **stronger and harder**, resisting further deformation.

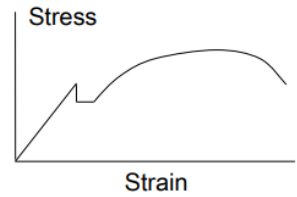
Real metals are not perfectly plastic

- They exhibit **work hardening** (also called strain hardening), which **limits how much they can be deformed**.

Materials are not perfectly plastic.



Real metals exhibit work hardening.



- This limits the extent of deformation possible, but it also increases the hardness of the material which can be desirable.
- This is the basis for the difference between cold working and hot working metals.
- In cold worked metals, the product is stiff, hard and strong.
- However the forces necessary to deform the metal are high.
 - This can also result in in-built stresses and possible problems with corrosion in service.

Work hardening increases hardness

- This can be **useful** for strength, but also makes further shaping harder.

Cold working vs. Hot working

- This difference in behavior underpins the **contrast**:
 - **Cold working** happens below recrystallization → strong but difficult to deform
 - **Hot working** allows easier shaping with less hardening

Cold worked metal = Stiff, hard, strong AND Higher forces are needed to deform hardened metal

SHEET DEFORMATION

Process	Description	Key Stress State	Notes
Blanking	Cutting sheet using punch & die	Shear	Removes a shape from the sheet
Deep Drawing	Forming sheet into a cup or can	Biaxial/Triaxial	Involves bending and stretching
Limitation	Side walls may fracture	Tensile stress	Walls thicken; overstretching causes failure

Deep Drawing:

- **No volume change:**

The material doesn't lose volume, so **side walls may thicken** as the material is drawn inward.

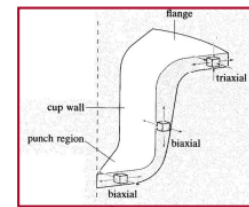
- **Stress conditions:**

- Most of the process involves **biaxial stresses** (stretching in two directions).
- Some regions (like the flange) experience **triaxial stress** (important in fracture behavior).

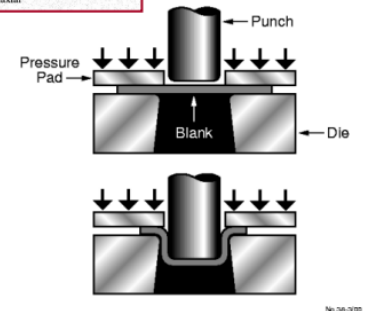
- **Deformation Limits:**

- Limited by the ability of the side walls to carry the tensile loads.
- Eventually, if overstretched, the walls **fracture or tear**.

- Blanking (cutting via shear forces punch and die)
- Deep drawing - bending, reverse bending.
- No change in volume and hence the side walls may **INCREASE** in thickness.
- Deformation limited due to side wall carrying forces which will ultimately fracture.



- Drawing usually consists of only biaxial stresses.



HOT WORKING

Hot worked metals can exhibit 2 features:

- REDUCED YIELD STRESS
- RECOVERY AND RECRYSTALLISATION

MACHINING

Machining refers to processes where **material is removed** (cut, ground, or milled) from a workpiece to shape it with high precision. Examples include:

- **Turning, milling, drilling, grinding, and CNC machining**

Often a secondary operation to 'finish' parts produced by e.g., casting.

Sometimes it is PRIMARY process:

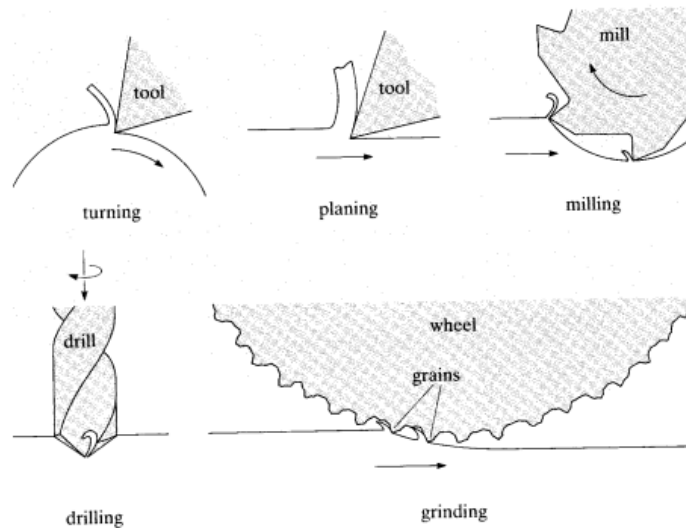
- **Low-volume production**
- **Prototyping**

Doesn't need dedicated tooling for each part — **generic machines like CNC mills** or lathes can handle various designs.

Tooling costs = LOW!

CUTTING PROCESSES

- General requirement for cutting operations is that cutting tool is harder than work material and work piece is ductile.
- Single point cutting operations include:
 - turning,
 - planing,
 - milling,
 - drilling, and
 - grinding.



JOINING – METAL FABRICATION METHODS

JOINING

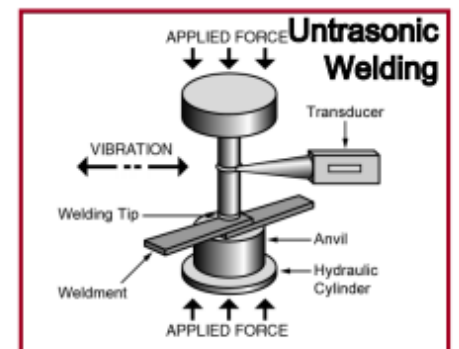
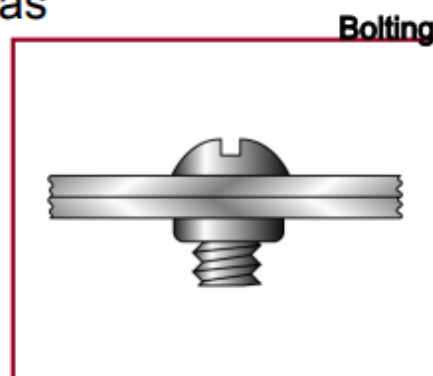
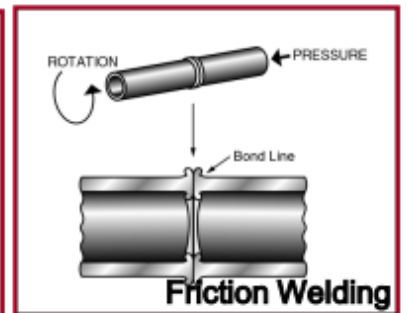
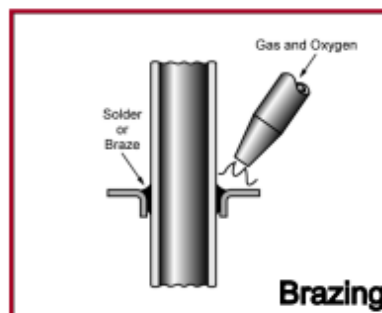
- Joining can involve:

- Welding
- Fastening
- Adhesive bonding

- Some facilitate disassembly such as using:

- bolts, and
- screws.

- Others do not.



- Uses **heat and a filler metal** (like solder) to **join parts without melting the base metal**.
- Gas torch shown adding filler between closely fit metal pieces.
- **Moderate strength**, commonly used in pipes and electronics.

Friction Welding

- Parts are **rotated and pressed together**.
- Friction **heats the interface**, and pressure **bonds them** without melting.
- Often used in automotive/aerospace shafts.

Ultrasonic Welding

- A **vibrating tool (sonotrode)** applies **high-frequency mechanical vibrations** with pressure.
- Used for **plastics or light metals** (e.g., copper wires, lithium-ion batteries).
- Fast, clean, and **no filler needed**.

Bolting

- **Mechanical fastener** (bolt + nut or threaded hole).
- **Easy to assemble/disassemble**.
- Common in construction, automotive, machinery.

Phase	Properties	Use in steel
Pearlite	Strong but softer than martensite	Ductile & used in mild steels
Martensite	Extremely hard & brittle	☒ Main strengthening phase
Bainite	Harder than pearlite, softer than martensite	Good compromise phase
Austenite	Only stable at high temperature	Not strong, used before quenching

What Quenching *Does* Change:

- **Strength (Yield & Ultimate)**: increases due to formation of **martensite**.
- **Hardness**: increases dramatically.
- **Toughness**: can decrease if material becomes too brittle.
- **Ductility**: usually decreases.

Grain Size	Strength	Ductility	Common In
Small	High	Low	Aircraft parts, tools
Large	Low	High	Sheet forming, wires

4. 4. What is the purpose of normalizing steel?

- a) Increase hardness
- b) Refine grain size
- c) Produce martensite
- d) Improve machinability

Answer: b) Refine grain size

links with annealing pic! Refine grain size!

9. 9. In work-hardened metals, which property increases?

- a) Ductility
- b) Electrical conductivity
- c) Yield strength
- d) Grain size

Answer: c) Yield strength

10. 10. Spheroidizing is done in steels to:

- a) Improve corrosion resistance
- b) Increase ductility
- c) Increase hardness
- d) Promote carbide formation

- Makes very soft steel therefore increase ductility. DUCTILITY = able to bend and stretch without breaking A **ductile material** can undergo a lot of plastic deformation before it fractures.
- Think of it like this: **You can bend, stretch, or draw it into a wire without it snapping.**

15. 15. In the TTT diagram, the vertical path from high to low temperature represents:

- a) Slow cooling
- b) Annealing
- c) Air cooling
- d) Quenching

Answer: d) Quenching

THIS DIAGRAM SHOWS THE HEAT TREATMENT

1. Austenitizing (Heating)

- The metal (usually steel) is heated to a high temperature (around 800–950°C).
- At this stage, the steel transforms into a single-phase structure called **austenite**.

🔧 Purpose: Prepare the steel's structure for transformation during cooling.

Then:

a) Annealing (Blue Path)

- Slow cooling** from the austenite region.
- Follows the **right side of the curve**, transforming to **pearlite or ferrite** over time.
- Grain growth and softness** occur.

✅ Results:

- Low strength**
- High ductility**
- Soft and machinable**

b) Quenching (Red Path)

- Rapid cooling** (steep vertical drop).
- Bypasses the “nose” of the TTT curve to **avoid diffusion-based transformations**.
- Forms **martensite**, a **hard and brittle** phase.

✅ Results:

- Very high strength and hardness**
- ✗ **Very low toughness**

c) Tempered Martensite (Pink Path)

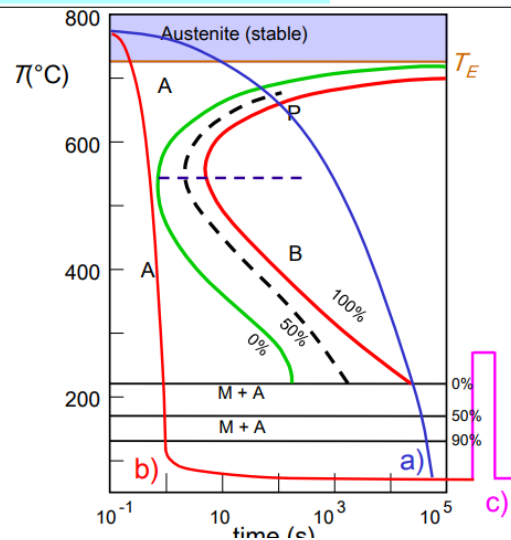
- First, quench → forms **martensite**

Heat Treatments

a) Annealing

b) Quenching

c) Tempered Martensite



- Then **reheat gently** to around 200–600°C.
- This tempers the martensite (horizontal pink step).

✅ Results:

- **Slightly reduced strength**
- ✅ **Greatly improved toughness**
- ✅ **More ductile and usable**

Zone	Meaning
A	Austenite region (high temp, single phase)
M + A	Mixed Martensite and Austenite
Region P to B	Diffusion-controlled transformations like bainite/pearlite
Martensite line (below 200°C)	Rapid transformation to martensite if cooled fast enough

Steel Heat Treatment Phases — Simple Explainer

1. Austenite (γ)

- **When?** At high temperature (~800°C)
 - **What is it?** A solid solution of carbon in face-centered cubic (FCC) iron.
 - **Important because?** It's the parent phase. All other phases come from it depending on how you cool it.
-

2. Pearlite

- **When?** Forms if you **cool slowly**
 - **How?** Carbon atoms **have time to diffuse** out, forming alternating layers (lamellae) of:
 - **Ferrite (soft) + Cementite (hard)**
 - **Properties:** Strong and ductile, but **not very hard**
 - **Seen in:** Normal air-cooled steel
-

3. Martensite

- **When?** Forms if you **quench (rapidly cool)**
- **How?** Cooling is so fast that carbon **can't diffuse**, so atoms are "trapped"
- **What happens?**
 - The crystal distorts into a **Body-Centered Tetragonal (BCT)** shape
- **Properties:** **Very hard and brittle**
- **Key fact:** It is a **diffusionless and very rapid** transformation

- **Seen in:** Tool steels, hardened blades

4. Tempered Martensite

- **When?** You take martensite and **reheat it slightly**
- **Why?** To reduce brittleness and improve toughness
- **Used in:** Spring steels, knives, structural parts

Cooling Rate	Phase Formed	Process	Properties
Slow (air)	Pearlite	Annealing	Strong, ductile
Medium (oil)	Bainite	Normalizing	Moderate strength/toughness
Rapid (water)	Martensite	Quenching	Very hard, brittle
Martensite + heat	Tempered Martensite	Tempering	Tough + hard (good balance)

Phase	Part of Heat Treatment?	How It Forms
Austenite (γ)	YES (starting phase)	Formed by heating steel to $\sim 800^{\circ}\text{C}$
Pearlite	YES (after annealing)	Slow cooling $\rightarrow$ diffusion allowed
Martensite	YES (after quenching)	Rapid cooling $\rightarrow$ diffusion blocked
Tempered Martensite	YES (after tempering)	Gentle reheating of martensite

Summary Table: Metal Processing Techniques

Technique	Type	Key Features / Purpose	Notes / Examples
Sand Casting	Casting	Uses expendable sand moulds	Good for large parts, low cost, rough finish
Die Casting	Casting	Uses reusable metal moulds	High precision, for low-melting alloys
Investment Casting	Casting	Expendable wax pattern and ceramic shell	High accuracy, complex shapes
Continuous Casting	Casting	Solidifies metal into sheet/bar directly	Used in steel industry, avoids contraction defects
Forging	Forming	Deforms metal using compressive force	Improves grain flow, strong parts
Drawing	Cold Forming	Pulls metal through die to reduce size	Used for wires, tubing

Rolling	Forming	Compresses metal between rollers	Hot or cold, makes sheets and I-beams
Extrusion	Forming	Forces metal through die	Rod/tube shapes; hot or cold
Machining	Secondary / Primary	Cuts metal to shape with tools	CNC, milling, drilling; good finish
Powder Metallurgy	Forming	Compacts metal powders and sinters	For high-melting metals like tungsten
Brazing	Joining	Fills joint with molten filler metal	No melting of base parts
Welding	Joining	Fuses parts by melting	Strong joint, not removable
Bolting/Fastening	Joining	Mechanical attachment	Removable, uses screws/bolts
Annealing	Heat Treatment	Heat then slow cool	Softens metal, refines grains
Quenching	Heat Treatment	Rapid cooling from high temp	Forms hard martensite
Tempering	Heat Treatment	Reheats quenched steel slightly	Reduces brittleness of martensite
Normalizing	Heat Treatment	Air cool from above critical temp	Refines grain structure
Spheroidizing	Heat Treatment	Hold just below critical temp	Increases ductility for machining

SPL 6 – CERAMICS

Ceramics are: = non-metallic compounds made from metal + non-metal elements.

- **Inorganic & non-metallic materials.**

INORGANIC - made from **natural elements** like **silicon, oxygen, aluminum** - **Not made of carbon chains (unlike organic materials like plastics or wood)**

- Made from metal + non-metal elements.
- Bonding is ionic or covalent with some ionic.
- Name comes from Greek "*keramikos*" (pottery, "to burn").

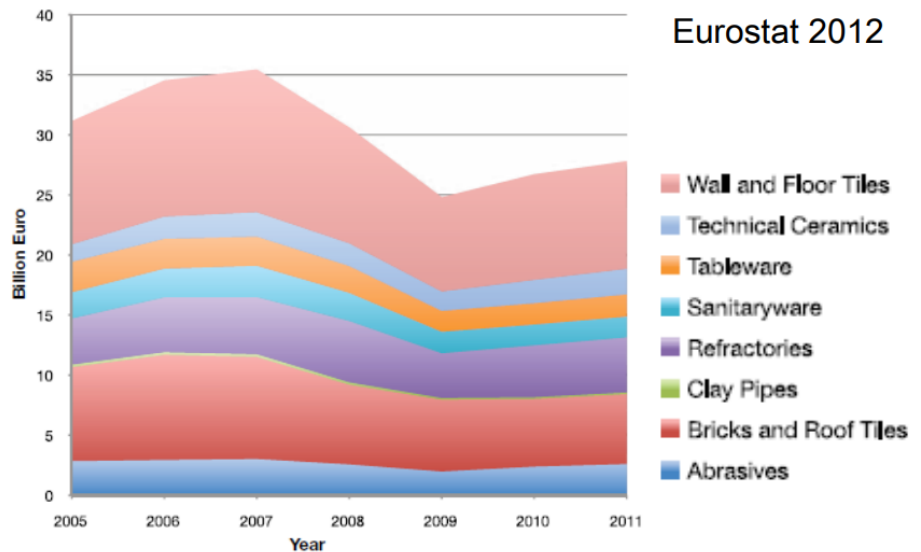
Traditional Ceramics:

- Used before 1950, made mainly from clay.
- Includes china, porcelain, bricks, tiles, glass.
- After 1950: new materials and better understanding → advanced ceramics.
- Now "ceramic" = a broad category of materials.

European Ceramics Production (2005–2011)

-  **Production peaked in 2007 (~35 billion euros), then dropped sharply around 2009** — likely due to the global financial crisis.
-  **Wall & floor tiles and bricks/roof tiles dominate the market.**
-  **Technical ceramics and abrasives make up a smaller portion.**
-  **Recovery starts after 2009 but doesn't reach pre-crisis levels by 2011.**

European Ceramics Annual Production



MAT602 – Materials Selection in Design.

Modern Technical Ceramics

Recent Materials Advances: Improvements in ceramic materials have made them essential in modern engineering due to their enhanced properties.

Wear and Corrosion Resistance: Ceramics are highly resistant to wear and corrosion, making them ideal for harsh environments such as in **pumps, nozzles, and valve parts** where durability is critical.

Thermal Properties: Ceramics can withstand high temperatures and resist thermal shock (sudden temperature changes), which is valuable for **automotive engine components** where materials must endure intense heat and rapid cooling.

Precision Manufacturing: Modern techniques allow ceramic parts to be made with **extreme precision**, with tolerances within a fraction of a micron. This makes them suitable for **high-tech and delicate applications** like electronics and aerospace.

Ceramic Material Groups

1. Structural Ceramics -used in high temp applications

- Used where **strength + heat resistance** are needed.
- Examples: **engine parts, cutting tools, furnace linings.**
- Materials:

- **Si₃N₄ (silicon nitride)**
- **Al₂O₃ (alumina)**
- **ZrO₂ (zirconia)** — tough and strong at high temps.

2. Electrical Ceramics

- Used in **electronics and magnets**.
- They can conduct or store electricity in special ways.

KEY APPLICATIONS:

- **Capacitors** – to store electrical charge
- **Sensors** – especially in medical or ultrasonic devices
- **Piezoelectric components** – generate voltage when squeezed (used in microphones, ultrasound, ignition systems)
- Examples:
 - **PZT** (used in sensors & ultrasound)
 - **BaTiO₃** (a capacitor material)

3. Bioceramics

- Used in the **human body** — safe and non-reactive.
- Examples:
 - **Al₂O₃** for hip joints
 - **Hydroxyapatite** mimics bone
 - **Glass** used in dental and bone repair

Seals and industrial machine parts

Why ceramics are used here:

- **Applications need:**
 - **Wear resistance** (must not degrade easily with use)
 - **High surface finish** (smooth, precise surfaces)
 - **Low friction** (so parts move easily without damage)
 - **VERY HIGH TEMP THESE MATERIALS NEED TO HAVE HIGH STRENGTH !**

Ceramics used:

- **Alumina (Al₂O₃)** – common, strong, corrosion-resistant
- **Silicon Carbide (SiC)** – extremely hard, good at resisting wear
- **Silicon Nitride (Si₃N₄)** – strong, tough, and handles shocks

Key ceramic properties:

- **Hardness** – resists scratches and deformation
- **Low porosity** – fewer tiny holes means stronger and smoother

- **Machinability to high finish** – they can be shaped and polished very precisely
- **Corrosion and wear resistance** – perfect for harsh machine environments

TURBINE STATOR

A **turbine stator** is a **stationary part** of a turbine (like in a jet engine or power plant - It directs high-speed hot gas **onto the rotating blades**

Application Requirements:

Turbine stators face extreme environments, so the materials must handle:

- **Thermal shock resistance:** sudden temperature changes without cracking
- **Oxidation resistance:** avoid reacting with air at high temperatures
- **High-temp stability:** maintain shape and strength when very hot
- **Complex aerodynamic shape:** must be precision-shaped to guide airflow

fabricated to complex shape



Candidate Ceramics:

- **Si₃N₄ (Silicon Nitride)**
- **SiC (Silicon Carbide)**
→ These are **advanced ceramics** known for high heat and wear resistance.

Key Properties:

- **High strength:** withstands stress and load
- **Moderate to low thermal expansion:** resists size change when hot (important for tight fits)
- **Good thermal conductivity:** can manage heat without overheating
- **Fabricatable into complex shapes:** essential for aerodynamic designs

HEATING ELEMENT

Requirements of the application:

- Must **generate heat** when electricity flows.
- Needs to survive **high temperature** and repeated **thermal cycling** (heating and cooling again and again).

Candidate Ceramics:

MoSi₂ (Molybdenum disilicide) SiC (Silicon carbide) C (Carbon), ZrO₂ (Zirconia) – doped to modify its electrical/thermal behavior.

Key Properties:

- Needs **semiconductor-like resistivity**:
 - Not a perfect conductor (like copper),
 - Not a perfect insulator either.
 - This balance lets it **generate heat** efficiently.
- Also must **resist oxidation** and remain stable in furnace conditions.

Why Semiconductor?

- Semiconductors allow **controlled resistance**:
Higher resistance → **More heat** (Joule heating)
But low enough to still **let current flow**.
- Metals would overheat or oxidize too quickly.
- Semiconducting ceramics handle **extreme temps** without breaking down.

HEAT SINK FOR TRANSISTOR DEVICES

A heat sink is a component used to **pull heat away** from electronic parts like **transistors** so they don't overheat.

A **transistor** is a tiny electronic component that acts like a **switch** or an **amplifier** for electrical signals.

Aspect	Details
Purpose	Dissipate heat from transistor devices (e.g., Si, GaAs chips)
Application Needs	- Rapid heat dissipation- Low electrical loss- Compatibility with chips
Candidate Ceramics	- BeO (Beryllium Oxide)- AlN (Aluminium Nitride)- Diamond
Key Properties	- High thermal conductivity (fast heat transfer)- High electrical resistance (electrical insulation)- Low dielectric constant (suitable for high-frequency circuits)
Where found in	Computers & laptops – especially CPUs and GPUs

FURNACE INSULATION

Category	Details
Requirements of the application	Withstand high temperatures, retain heat within furnace
Candidate ceramics	Porous fire bricks, fiberboard, 'wool' insulate furnace
Key properties	Low thermal conductivity, high-temperature stability

MINIATURE CAPACITOR

A **miniature capacitor** is a small electrical component used to **store and release electrical energy** in electronic circuits.

Think of it like a tiny battery, but it charges and discharges very quickly

- High dielectric constant → more charge stored in a small size
- Good temperature stability

- Low loss → efficient performance

Aspect	Details
Application Requirement	High charge-storage capability
Candidate Ceramics	Various titanates and mixed zirconate-titanates
Key Properties	High polarizability, high dielectric constant, low dielectric loss

Feature	Purpose/Benefit
High charge storage	Needed for compact, powerful electronics
Titanate ceramics	Good electrical properties for capacitor function
High polarizability	Enhances electric field interaction
High dielectric constant	Allows more charge storage in small volume
Low dielectric loss	Keeps energy loss and heat generation minimal

HIGH SPEED, HIGH LOAD BEARING

A high-speed, high-load bearing is a **mechanical component** that:

- Supports rotating or moving parts (like a shaft or wheel)
- Withstands very heavy loads
- Operates at high speeds without failing

APPLICATION REQUIREMENTS:

- **High contact loads:** Must withstand heavy pressure.
- **Wear resistance:** Should not degrade quickly.
- **Low friction:** Important for speed and efficiency.
- **Heat dissipation:** Must release heat fast to avoid damage.

Category	Details
Application Requirements	- Resistance to high contact loads- Wear resistance- Low friction- Heat dissipation
Candidate Ceramics	- Silicon Nitride (Si ₃ N ₄)
Key Properties	- High toughness and strength (resists breakage under load)- High hardness (resistant to wear)- Very low porosity (fewer internal flaws, better reliability)- Machinable to high surface finish (smooth = less friction)- Moderate to high thermal conductivity (helps carry away heat during use)
Use Case Summary	Ideal for extreme mechanical and thermal environments like aircraft engines, racing cars, and precision machines. Combines strength, precision, and durability .

TRANSDUCER

A **transducer** is a device that **converts one form of energy into another**. In this case, it converts **mechanical movement (displacement or pressure)** into an **electrical signal**.

Element	Explanation
Requirements	Detect tiny movements (like pressure or vibration) and turn them into voltage.
Candidate Ceramic	PZT (Lead Zirconate Titanate) – a piezoelectric ceramic .
Working principle	When PZT is squashed or stretched , it generates voltage – just like how old weighing scales used to work.
Diagram (right) check ppt	Shows a block of PZT. When pressed (force applied), the gauge detects electricity produced from the deformation.

CUTTING TOOLS

● Requirements of the applications:

Ceramic grades run at high speeds and feeds on a variety of work piece materials.

Typical ceramic applications include **turning, grooving, and machining** of tough materials like hardened steels.

● Candidate ceramics:

ZrO₂ (Zirconia), WC (Tungsten Carbide)

● Key properties:

Zirconia ceramic has the **highest strength and toughness** at room temperature of all engineering ceramics.

- High hardness (for cutting tough materials)
- High toughness (resists cracking or breaking)
- Good thermal stability (can handle heat from friction)

Drill bit – used in hand drills or machines to make holes in metal or wood.

Other examples: **Lathe tool insert, End mill cutter AND Ceramic scissors** (like shown in the slide)

Property	Why It's Useful in Cutting Tools
High strength	Allows the tool to resist breaking under pressure.
Toughness	Withstands shock loads without cracking.
Hardness	Helps cut even tough materials like steel.

GRINDING MILL PARTS

Requirements of the applications:

Grinding mills are used to crush and grind solid materials into smaller particles. These machines require components that can **withstand very high wear and friction**.

That's why ceramics are used for parts like:

- Liners (interior walls)
- Agitator screws
- Rollers

- **Ball mill balls:**

The balls inside a **ball mill** (a type of grinder) are made from the **same ceramic material** as what's being ground. This avoids contamination.

What's a mill?

A *mill* is a machine that breaks down solids into finer particles.

- **Grinding mill** = breaks/crushes things
- **Ball mill** = filled with ceramic/metal balls that tumble and crush material

NOZZLES

Aspect	Details
Application Requirements	- Chemical and wear resistance (scrubber nozzles) - Withstand high-speed fluid flow (cleaning nozzles) thus wear resistance is essential - CERAMICS High heat resistance used in- (welding/burner nozzles)
Reason Ceramics Are Used	- Ceramics provide excellent wear and chemical resistance - Handle abrasive and hot conditions without degrading
Examples of Use	- Scrubber systems in chemical plants - High-pressure water jets - Welding/burner nozzles (extreme heat)

MEDICAL EQUIPEMENT AND CHEMICAL ANALYSER PARTS

- The chemical stability of ceramics permits their application to the artificial kidney pump.
- Free from corrosion and superior in cleaning properties, ceramic materials are also used in blood valves.

Aspect	Details
Application	Artificial kidney pumps, blood valves, chemical analyser parts
Reason ceramics are used	- Chemically stable (won't degrade in harsh environments) - Corrosion-resistant - Easy to clean (important for medical hygiene)
Key benefit	Safe for internal medical devices due to bio-inertness and cleanliness

FOOD PROCESSING MACHINNE PARTS

Aspect	Details
Application	Food processing machines – kneading rollers, valves, pumps
Reason ceramics are used	Ceramics are inherently clean and easy to sterilize
Key Benefit	Simplifies hygiene and cleaning procedures – essential in food industry

- Ceramics don't corrode or react with food ingredients, which makes them safe and durable for repeated food contact.

- Their smooth surface reduces bacterial buildup.

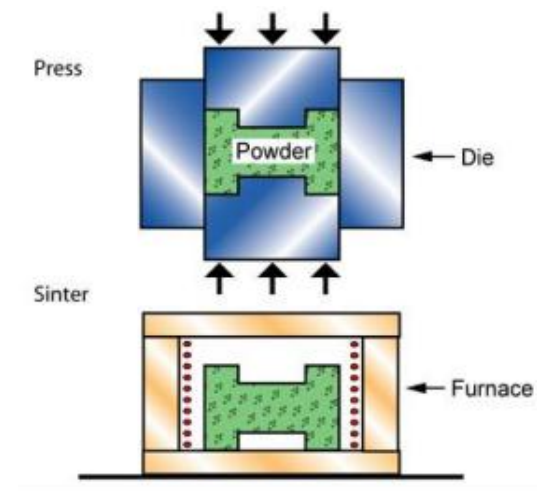
ISUZU ENGINE

An **Isuzu engine** is an internal combustion engine made by **Isuzu Motors**, a Japanese company GAS TURBINE!!

Aspect	Details
Application	Engine components (e.g. gas turbines, glow plugs)
Why ceramics are used	Excellent thermal efficiency , low energy loss , and heat resistance use of ceramic glow plug - Traditional ones are metal, but ceramic glow plugs heat up faster , reach higher temperatures , and are more durable —this improves engine efficiency and reduces emissions .
Key benefits	Clean energy consumption, withstands high temperatures, durable in engine environments

CERAMIC PROCESSING

Step	Explanation
1. Milling	Raw ceramic powders are mixed in distilled water and ground to reduce particle size.
2. Drying & Granulation	Water is removed; the powder is crushed and sieved to break agglomerates.
3. Compaction	The dried powder is placed in a die and pressed into shape (e.g., block, disc).
4. Sintering	The compacted part is heated in a furnace just below melting point. This bonds particles together and densifies the material.



Milling

- **What it means:** Breaking down raw ceramic materials into very fine powder.
- **Why it's done:** To reduce particle size and ensure uniformity for shaping and sintering.

◆ Agglomerated

- **What it means:** When fine powder particles stick together into clumps or clusters.
- **Why it matters:** These clumps must be broken (by **granulation**) to ensure even shaping and prevent defects.

◆ Sintering

- **What it means:** Heating the pressed powder (but not melting it!) to bond the particles together.

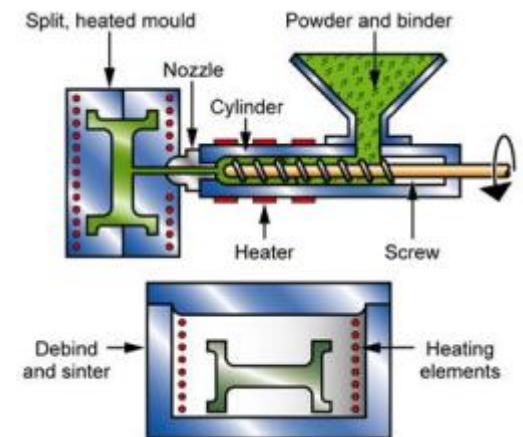
- **Why it's important:** It densifies the material, increases strength, and gives it final properties.

♦ Green Part

- **What it means:** The shaped ceramic component *before* sintering.
- **Think of it as:** A fragile, chalk-like version of the part that hasn't been fired yet.
- **Fun fact:** It's called "green" not because of color, but because it's not "finished" or fully processed yet.

CERAMIC INJECTION MOULDING

Step	Explanation
1. Mixing	Fine ceramic powder is combined with an organic binder (like plastic or wax).
2. Injection Moulding	The mix is injected into a mould using a screw/barrel system like in plastic injection moulding.
3. Binder Removal	The binder is removed by heating (thermal debinding) or with a solvent (chemical debinding).
4. Sintering	The 'green' part (now binder-free) is sintered to achieve full density and strength.



ADDITIVES

Additives are small amounts of extra substances added to ceramic powders to improve processing or final properties.

- **Additive choice depends on:**
 - Required *green strength* (how strong the part is before sintering).
 - *Machinability* – how easily it can be shaped.
 - *Compatibility* with the ceramic material.
 - The *type of consolidation process* (e.g., pressing, injection moulding).
- **Organic additives like waxes, gums, thermoplastic resins:**
 - Not water-soluble → not ideal for slip casting.
 - Good for **injection moulding** due to their meltability.
 - Can be **burned off** during sintering with minimal contamination.

- **Inorganic binders** remain part of the ceramic after processing.

Types of additives:

CERAMICS: STARTING POWDERS ARE DEPENDENT ON THE PROPERTIES REQUIRED IN THE FINISHED COMPONENT.

MUST BE PURE

MUST HAVE SMALL PARTICLES - SMALLER PARTICLES = MORE REACTIVE

Binder	Increase green strength and provide lubrication.
Lubricant	Mold releases, decrease particle-tool and inter-particle friction
Compaction aid	Aid in particle rearrangement during pressing.
Plasticizer	Rheological aid, improving flexibility of binder films.
Deflocculant	pH control, control of surface charge on particles, dispersion, or coagulation.
Wetting agent	Reduction of surface tension.
Water retention aid	Retain water during pressure application.
Antistatic agent	Charge control.
Antifoam agent	Prevent foam or bubble formation.
Foam stabilizer	Enhances foam formation.
Chelating agent	Deactivate undesirable ions.
Fungicide	Stabilize against degradation with aging.
Sintering aids	Aid in densification.
Dopants	Modify electrical, magnetic, optical properties.
Phase stabilizers	Control the crystalline phases present.

Purity in Ceramics – Summary

- **Purity greatly affects:**
 - Mechanical strength
 - Stress rupture life
 - Oxidation resistance
- **Impurity effects depend on:**
 - The type of impurity
 - How it's distributed
 - Operating conditions (temperature, stress, time)
- **Example:**
 - Calcium (Ca) reduces creep resistance in Si_3N_4 when pressed with MgO .
 - But **Ca** has little effect when Y_2O_3 is used instead of MgO .
- **For electrical, magnetic & optical ceramics:**
 - Even tiny changes in impurity levels (DOPANT) can **severely affect properties**.

Particle Size and Reactivity – Summary

- **Goal of consolidation:** Maximize packing & uniformity → reduce shrinkage & porosity after sintering.
- **Uniform size particles** pack poorly → ~30% void space.
- **Using mixed particle sizes** (big + small + smaller) improves packing:
 - 2 sizes → voids ↓ to 26%
 - 3 sizes → voids ↓ to 23%
- **Real ceramic particles** are irregular → porosity often >50% before sintering.
- **Low porosity & fine grains = high strength.**
- **For refractories** (where strength isn't key), large particles or high porosity may be beneficial for:
 - **Low thermal conductivity**
 - **High thermal shock resistance**

Reactivity Slide:

- **DENSIFICATION OF A COMPACTED POWDER AT HIGH TEMP** is driven by a change in **surface free energy** — particles bond to reduce this energy.
- **Smaller particles** (around **1 μm or less**) have **higher surface area thus high surface free energy therefore STRONGER THERMODYNAMIC DRIVE TO DECREASE THEIR AREA BY BONDING TOGETHER**. Making them more reactive and easier to **bond and densify**.
- For polycrystalline alumina for sodium vapour lamp envelopes to achieve transparent- **transparent ceramics** (like in sodium lamp envelopes), **pores bigger than 0.4 μm** must be removed during sintering, as they scatter visible light and reduce transparency.

● Small particles = more bonding

● No big pores = better clarity

SINTERED Si_3N_4

- **Larger particles ($\approx 2 \mu\text{m}$)** → Only reach **$\sim 90\%$** of theoretical density after sintering.
- **Smaller particles (sub-micron)** with **high surface area ($\geq 10 \text{ m}^2/\text{g}$)** → Achieve **$>95\%$** density.

Why it matters:

Higher density means **fewer pores**, which leads to **stronger, more reliable ceramics**. So, using **finer powders** improves the quality of the final ceramic component.

POWDER SIZING

- **Why size matters:** Particle size and distribution affect the **final properties** (e.g. strength, density, wear resistance).
- **Abrasives (wears down, smooth or clean surfaces eg sandpaper in ceramics - ALUMINA):** These materials need to come in **specific size ranges** for effective use.
- **Technique variety:** Many **powder synthesis and sizing methods** exist to produce the exact particle size distributions needed for different applications.

🔗 In short: Getting the **right particle sizes** is critical for the **performance of ceramics**, especially in abrasive or structural roles.

PARTICLE SIZE- Finer powders → better **packing, sintering, and final ceramic properties** like **strength and density**.

SINTERING

Sintering = Heating the compacted powder so particles bond and pores shrink or disappear, making the part denser and stronger.

To sinter, two things are needed:

1. **Transport mechanism** – material must move from one particle to another.
2. **Energy source** – usually heat, to drive that movement.

Mechanism	Description
-----------	-------------

Particle Size

- Mechanical Screening
- Elutriation
- Air classification
- Ball milling
- Attrition milling
- Vibratory milling
- Turbomilling
- Fluid energy milling
- Hammer milling
- Roll crushing
- Chemical Precipitation
- Sol-gel
- Liquid mix
- Decomposition
- Freeze drying
- Hot kerosene drying
- Calcining
- Rotary kiln
- Fluidized bed
- Combustion synthesis

*Finer
Particle
Size*

Vapour phase	Atoms move through gas/vapour between particles.
Solid state	Atoms move directly through solid contact (no melting).
Liquid phase	A small amount melts and helps other particles bond.
Reactive liquid	A liquid is formed during sintering via a chemical reaction, aiding bonding.

Process	Transport Mechanism	Driving Energy
Vapour phase	Evaporation and condensation	Differences in vapour pressure
Solid state	Diffusion	Differences in free energy
Liquid phase	Viscous flow and diffusion	Capillary pressure, surface energy
Reactive liquid	Viscous flow + solution precipitation	Capillary pressure, surface energy

VAPOUR PHASE SINTERING

“Movement from a positive radius of curvature to negative radius with a lower vapour pressure.”

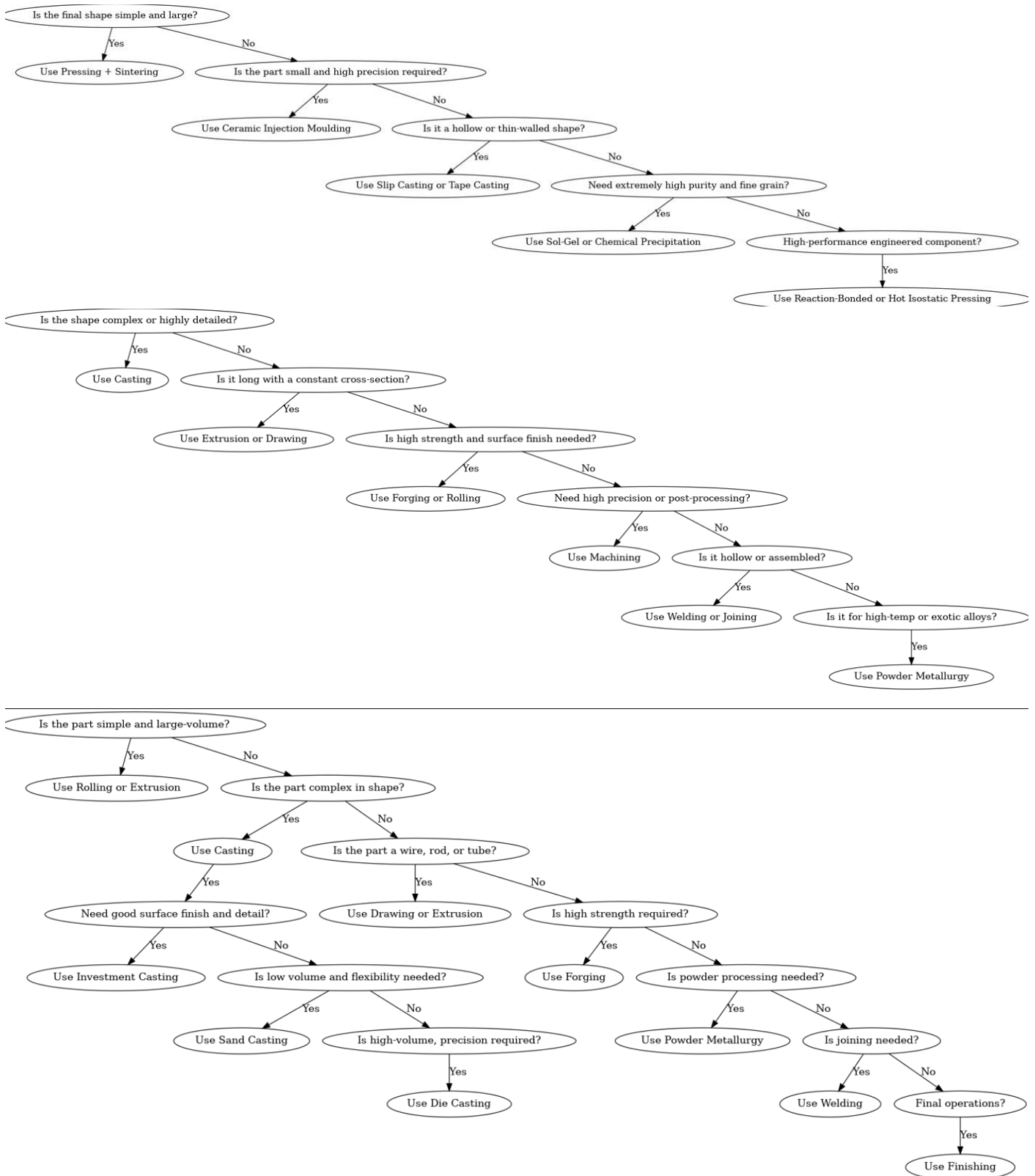
- **What it means:** Atoms evaporate more easily from **small curved surfaces** (positive curvature) and move to areas with **less curvature or indents** (negative curvature).
- **Why?** Smaller particles have **higher vapour pressure**, so atoms “leave” more easily.
- ◇ **“Smaller particles therefore have a stronger driving pressure.”**
 - Smaller particles = **higher surface energy** = **higher vapour pressure**.
 - So they **evaporate more readily**, driving material towards larger particles or neck regions.
- ◇ **“No shrinkage or densification.”**
 - Even though **material is moving**, the **overall structure doesn’t compact or shrink**.
 - Only **necks form** between particles, but **pores remain**, so the part stays porous.

SOLID STATE SINTERING

- **What happens?**
Particles **diffuse** into each other through the solid phase (they stay solid the whole time).
- **What drives it?**
A drop in **free energy** (particles want to reduce their surface area, like a balloon deflating).
- **Equation:**
The equation just models how far particles “sink into” each other over time.
- This **does cause densification**—particles come closer together, **pores shrink**, structure gets stronger.

Reactivity in Ceramic Sintering – Summary

- Particle size and reactivity determine how quickly and at what temperature sintering happens.
- Smaller particles (finer powders) have more surface area, so they sinter faster and at lower temperatures.
- **But:** Long sintering times can cause grain growth, which reduces strength.
- So to keep ceramics strong, you want a powder that can densify quickly with minimal grain growth.



For powder check slide!!